

Using rates 1A

RATES

UNIT 1
TOPIC 2

1

James is paid \$13.70 per hour.
How much is he paid for 7 hours' work?
Multiply the rate per hour by the number of hours:

James is paid _____.

2

Tomatoes were priced at \$5.90 per kilogram.
How much would 2.3 kg of tomatoes cost?

Multiply the rate per kilogram by
the number of kilograms:

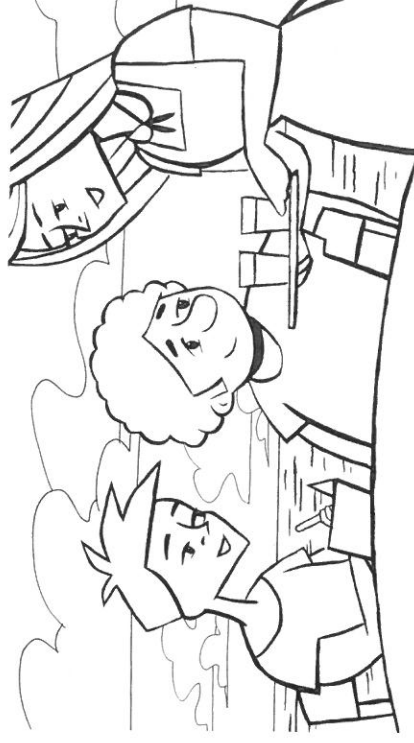
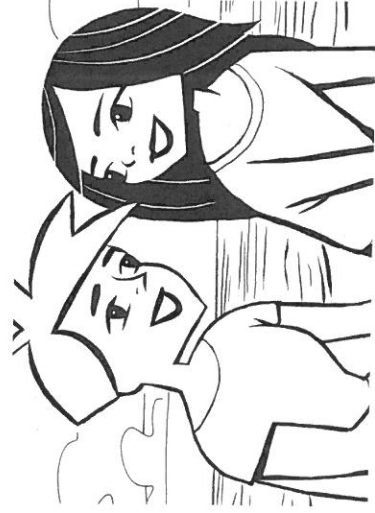
2.3 kg of tomatoes cost _____.

3

The instructions for a brand of fertiliser state that
1 kg should be applied to an area of 25 m².
(The rate of cover is 25 m²/kg.)
What area can be covered using 15 kg of this fertiliser?

Multiply the rate by the number of kilograms:

15 kg of fertiliser will
cover _____.



4

Rhani is paid \$14.50 per hour.
Calculate how much she earns when she works:

A

4 hours. _____

B

5 hours. _____

C

12 hours. _____

D

20 hours. _____

5

A tiler lays 2 m² of tiles per hour.
Find the number of square metres she can lay in:

A

3 hours. _____

B

5 hours. _____

C

1 $\frac{1}{2}$ hours. _____

D

3 $\frac{1}{2}$ hours. _____

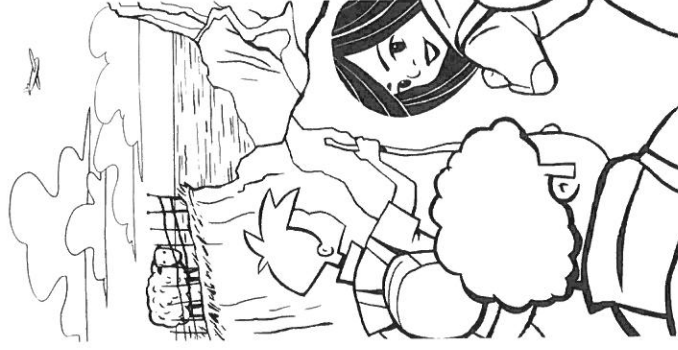
6 A cricket team scored an average of 5 runs per over.
How many runs did they score off 19 overs?

7 Rochelle can shear 14 sheep per hour.
How many sheep could she shear in 8 hours?

8 A farmer can have 10 sheep
per hectare on his property.
He has 75 hectares of land
for sheep.
How many sheep can he have?

9 An aeroplane can carry
135 passengers per flight.
Yesterday, it flew 8 full trips.
How many passengers
did it carry?

10 A heart is beating at 70 beats
per minute.
How many beats will it make
in 1 hour? (1 hour = 60 minutes)



11 Water is dripping from a tap at a rate of 4 litres per hour.
How much water will drip from the tap in 1 day?
(1 day = 24 hours)

12 Marko estimates that his feed supply will sustain
5 head of cattle per hectare.
How many head of cattle could he put on
an area of 250 hectares?

13 A mine supplies ore at a rate of 1 600 tonnes per day.
The mine operates 7 days each week.
How many tonnes of ore can it supply in 3 weeks?

14 The temperature at 8:00 am was 15°C.
If it increased at a steady rate
of 2°C per hour, what was
the temperature at:

- A** 9:00 am? _____
- B** 11:00 am? _____
- C** 12 noon? _____
- D** 10:30 am? _____
- E** 12:30 pm? _____



Using rates 2A

Many problems using rates can be solved using these steps:

- Step 1: Write down the rate.
 Step 2: Reduce the rate to 1 unit (by division).
 Step 3: Multiply or divide to get the answer.

To decide whether to multiply or divide, check the position of the required quantity in the rate.

- To find the first quantity you multiply.
- To find the second quantity, you divide.

1 Eleni delivers 320 letters in 4 hours.

A What is her delivery rate per hour? _____

B How many letters can she deliver in 5 hours? _____



2 David earns \$170.25 for 15 hours' work.

A How much is he paid per hour? _____

B How much would he earn for 7 hours' work? _____

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You may need to use working paper to answer these questions.



3 Sarah delivered 28 parcels in 4 hours.
 At this rate, how many parcels would she deliver in 5 hours?

4 A truck can carry 182 tonnes in 7 trips.
 How many tonnes can it carry in 5 trips?

5 On a sheep farm, 350 sheep can graze on 5 hectares of pasture.
 How many sheep can graze on 16 hectares?

6 A wheel spins at 45 revolutions per minute.
 How many revolutions will it make in 1 hour?

7 A factory fills soft drink bottles at a rate of 80 bottles per minute.
 How many bottles can be filled in 2 hours?

8 A pump can deliver 55 litres of water in 5 minutes.
 How many litres can it deliver in 3 hours?

9 A piece of fabric is 3 metres long. It costs \$48.

A What is the cost per metre? _____

B What would be the cost of a piece measuring 5 metres long? _____

C What length of fabric can be purchased for \$56? _____

10 Tyrone sheared 34 sheep in 2 hours. If he continues working at the same rate:

A what is Tyrone's shearing rate in sheep per hour? _____

B how long will it take him to shear 85 sheep? _____

11 Phuong can type at a rate of 75 words per minute.

How long will it take him to type a 1 500 word document at this rate? _____



12 A factory produces drink cans at a rate of 200 cans per minute.
How long would it take to produce 1 500 cans? _____

13 Martin serves an average of 24 customers every 2 hours.
At this rate, how long would it take him to serve 54 customers? _____

14 A team of bricklayers takes 6 hours to lay 1 500 bricks.
Working at the same rate, how long will it take them to lay 2 500 bricks? _____

15 The annual birth rate of a population is usually expressed as the number of births each year per 1 000 people.

A country has an annual birth rate of 24 per 1 000 people.
Calculate the number of births expected next year in a town of:

A 5 000 people. _____

B 30 000 people. _____

C 85 000 people. _____

D 165 000 people. _____

E 128 500 people. _____

